

Failure Mode Recognition Without Monitoring: An Anti-Crystallization Framework

M-ZRT, modified to reduce crystallization

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Step 1: Install Unreal Tournament or an equivalent game.

Step 2: Choose one exercise.

Step 3: Read the risk section of the selected exercise once, in order to understand what forms of monitoring, optimization, or behavioral capture the exercise may create.

Step 4: Resume playing and allow the procedure itself to recede.

The objective is to briefly recognize monitoring capture if it becomes obvious, then relax it rather than maintain it. Read the risk section once. Use it only to recognize obvious monitoring capture if it appears. Do not actively search for risks during the exercise, and do not check whether monitoring capture occurred correctly. Risk descriptions serve as one-time anti-crystallization cues rather than monitoring tools. They are intended to be read once and not actively recalled during task execution.

Path hesitation: Brief hesitation between two possible directions

risks:

Decision ritualization: you start manufacturing hesitation because "hesitation is the exercise."

Optimization contamination: hesitation becomes a tool, "I should hesitate now to reduce Z."

Excessive deliberation: instead of a tiny pause, it expands into actual comparison between paths.

Environmental scanning: you begin searching for locations that maximize ambiguity or produce "good" hesitations.

Meta-checking: monitoring whether the hesitation felt correct or long enough.

Action fragmentation: if overused, movement becomes choppy or unnatural because micro-decisions become foregrounded.

Binary fixation: repeatedly creating only left/right choices can crystallize a specific structure.

Question, no answer. Ask a question in your head (e.g., "What if I turned left?") but you do not attempt to answer it.

Risks:

Completion pressure: many questions automatically recruit answer generation. Even "What if I turned left?" can trigger route simulation, pros/cons, or prediction.

Meta-monitoring: you may start checking whether you successfully avoided answering. Then the exercise becomes performance evaluation.

Question collection: the brain may start generating many questions because the exercise itself becomes a search task, "find a good incomplete question."

Repetition crystallization: reusing the same question format repeatedly can create ritualization. For example, always using spatial questions.

Open loop persistence: some questions remain active and continue occupying working memory after the moment ends.

Goal contamination: the question becomes instrumentalized, "this question will lower Z" or "this question should create openness."

Exercise: Projectile as Phenomenon

Description: When a rocket, shock ball, or shot crosses space, it briefly becomes a visual event rather than just a threat. You continue playing normally.

Risks:

Can become active searching for visually satisfying projectiles.

May create anticipation loops: waiting for the "next good projectile."

The brain may begin preferring certain weapons purely for perceptual intensity.

Can drift into aestheticization of combat instead of embedded gameplay.

Repeated focus on projectiles may increase hypervigilance rather than openness.

The projectile can become a "special object" instead of a transient event.

The exercise may slowly detach from gameplay goals and become its own activity.

Can produce subtle self-monitoring: "Did I notice it correctly?"

The salience itself can become rewarding, leading to active pursuit of perceptual spikes.

Strong visual effects may encourage collecting memorable moments mentally.

High-intensity explosions or shock combos can become emotionally loaded anchors.

If repeated too intentionally, the exercise may turn into cinematic contemplation.

The player may unconsciously slow gameplay to preserve perceptual moments longer.

Certain maps or weapons may become associated with "better salience experiences," recreating optimization.

The transient nature of the projectile protects against fixation, but conscious repetition can still stabilize the pattern.